Fragile States Index Analysis

ITCS 6162 Knowledge Discovery in Databases

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Team Members

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1. Problem Description

State fragility describes the extent to which a government fails to provide its population with security, economic stability, political legitimacy, and basic public services. Fragile states are characterized by weak institutions, persistent conflict, and widespread poverty. They account for a disproportionate share of global humanitarian crises, forced displacement, and human rights violations.

The Fragile States Index (FSI), published annually by the Fund for Peace, quantifies fragility across 178–179 countries using 12 sub-indicators spanning three domains: Cohesion (C1–C3), Economic (E1–E3), and Political/Social (P1–P3, S1–S2, X1). A higher FSI score means greater instability. The FSI total score is the decision attribute in this project and is discretized into 9 ordinal categories following the Wikipedia FSI classification scheme.

This project answers two questions: (1) What combination of factors best predicts a country's fragility classification? (2) What actionable changes in those factors could move a country to a less fragile category? To answer these, we merged three years of FSI data (2019–2021), added four World Bank features, ran three classifiers (Decision Tree, Random Forest, Naive Bayes) in Python and J48 in WEKA, and extracted action rules using LISp-Miner's Ac4ft-Miner module.

2. Choice and Justification of Additional Features

Four features were added from the World Bank Open Data API. Selection was based on domain knowledge about fragility drivers and validated through Pearson correlation analysis.

Feature	World Bank Code	Type	Correlation with FSI
GDP per Capita (USD)	NY.GDP.PCAP.CD	Continuous	−0.779
Life Expectancy (years)	SP.DYN.LE00.IN	Continuous	−0.793
Internet Access (% pop.)	IT.NET.USER.ZS	Continuous	−0.722
Infant Mortality (per 1000)	SP.DYN.IMRT.IN	Continuous	+0.723

GDP per Capita: Economic poverty is one of the most established drivers of fragility. Countries with low income lack fiscal resources to maintain security forces, deliver services, and build resilient institutions. Correlation with FSI: −0.779.

Life Expectancy: Reflects cumulative quality of healthcare, nutrition, and absence of conflict. The strongest correlate of FSI at −0.793. Countries in civil war or extreme poverty show dramatically lower life expectancy.

Internet Access: A broad proxy for infrastructure development, education, and government transparency. Captures the development gap between stable and fragile states without overlapping the existing 12 sub-indicators. Correlation: −0.722.

Infant Mortality: Measures a government's ability to protect its most vulnerable citizens through healthcare access. Positive correlation of +0.723 means higher infant mortality consistently accompanies higher fragility.

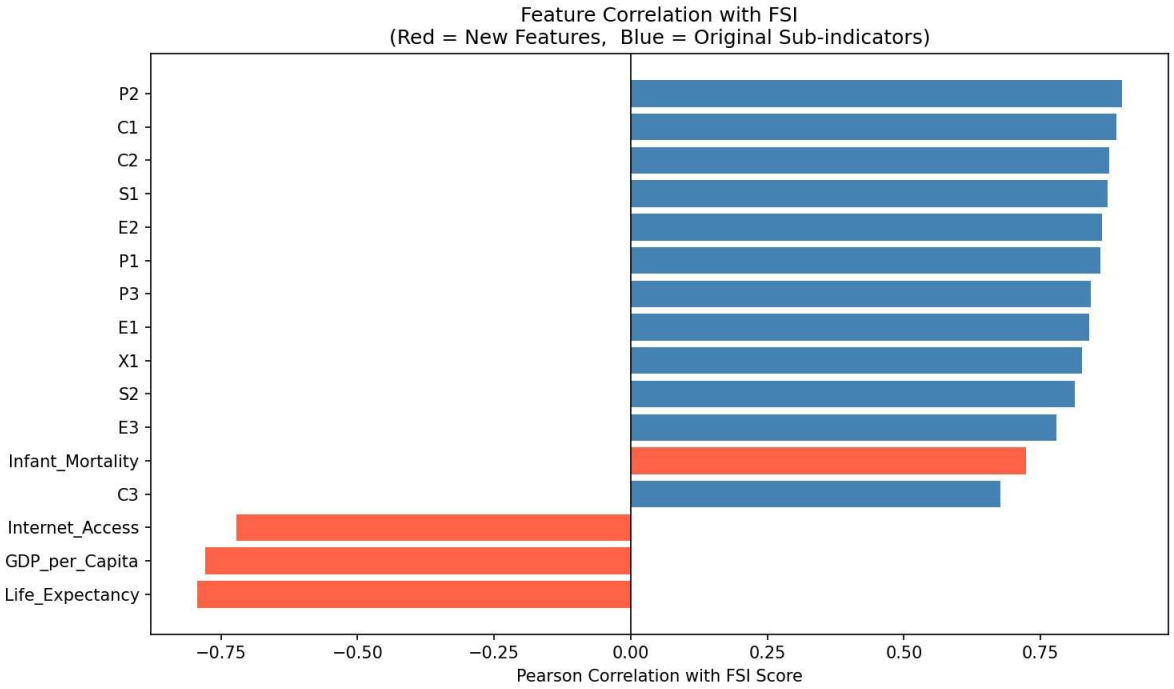

Figure 1: Pearson correlation of all 16 features with FSI score. Red = new World Bank features, Blue = original FSI sub-indicators.

3. Data Preprocessing

Step 1 Merging Annual Files: Three FSI Excel files (2019, 2020, 2021) were loaded and concatenated into a single DataFrame of 535 rows. A Year column was added. The “Change from Previous Year” column was dropped as it only existed in two of the three files.

Step 2 Country Name Standardization: FSI country names were mapped to World Bank format using a custom dictionary (e.g., “DR Congo” → “Congo, Dem. Rep.”) to enable successful API matching.

Step 3 World Bank API Extraction: Four features were fetched via the World Bank REST API for each country-year pair. JSON responses were parsed and merged by country and year.

Step 4 Missing Value Imputation: Approximately 8% of records had missing values for the new features. These were filled using median imputation stratified by year.

Step 5 FSI Score Discretization: The continuous FSI score was converted to 9 ordinal categories using the official Fund for Peace classification scheme documented on Wikipedia.

Step 6 Feature Discretization for LISp-Miner: All 16 features were discretized into Low/Medium/High using equal-frequency binning to produce fsi_lisp_miner.csv for LISp-Miner and fsi_weka.csv for WEKA.

Step 7 Train-Test Split: For Python classifiers, an 80/20 stratified split was used. WEKA used 10-fold cross-validation.

4. Classification Results and Analysis

4.1 Python Classifiers (scikit-learn)

Three classifiers were trained in Python with and without the 4 additional features:

Classifier	Accuracy (12 Features)	Accuracy (16 Features)	Change
Decision Tree	65.4%	62.6%	−2.8%
Random Forest	75.7%	79.4%	+3.7% ✓ Best
Naive Bayes	71.9%	68.2%	−3.7%

Random Forest achieved the best accuracy at 79.4% with the enriched feature set. The 3.7% improvement confirms that GDP per Capita, Life Expectancy, Internet Access, and Infant Mortality add meaningful predictive signal. Decision Tree and Naive Bayes showed slight degradation Decision Trees are prone to overfitting in higher-dimensional spaces, while Naive Bayes suffers when its conditional independence assumption is violated by correlated features.

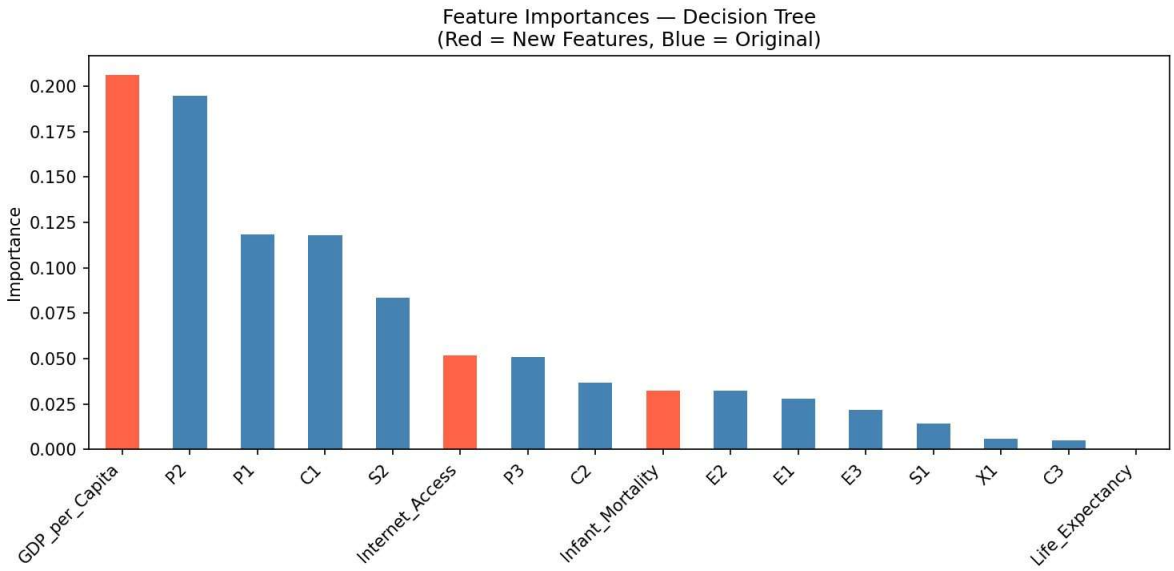

Figure 2: Decision Tree feature importances. GDP per Capita (red, new feature) ranks #1 at ~0.207, followed by P2 at ~0.195.

4.2 WEKA J48 Classification Step-by-Step Screenshots

Classification rules were also extracted using WEKA's J48 decision tree algorithm (equivalent to C4.5). The discretized dataset fsi_weka.csv (535 instances, 18 attributes) was used with 10-fold cross-validation.

Step 1: Loading fsi_weka.csv into WEKA Explorer

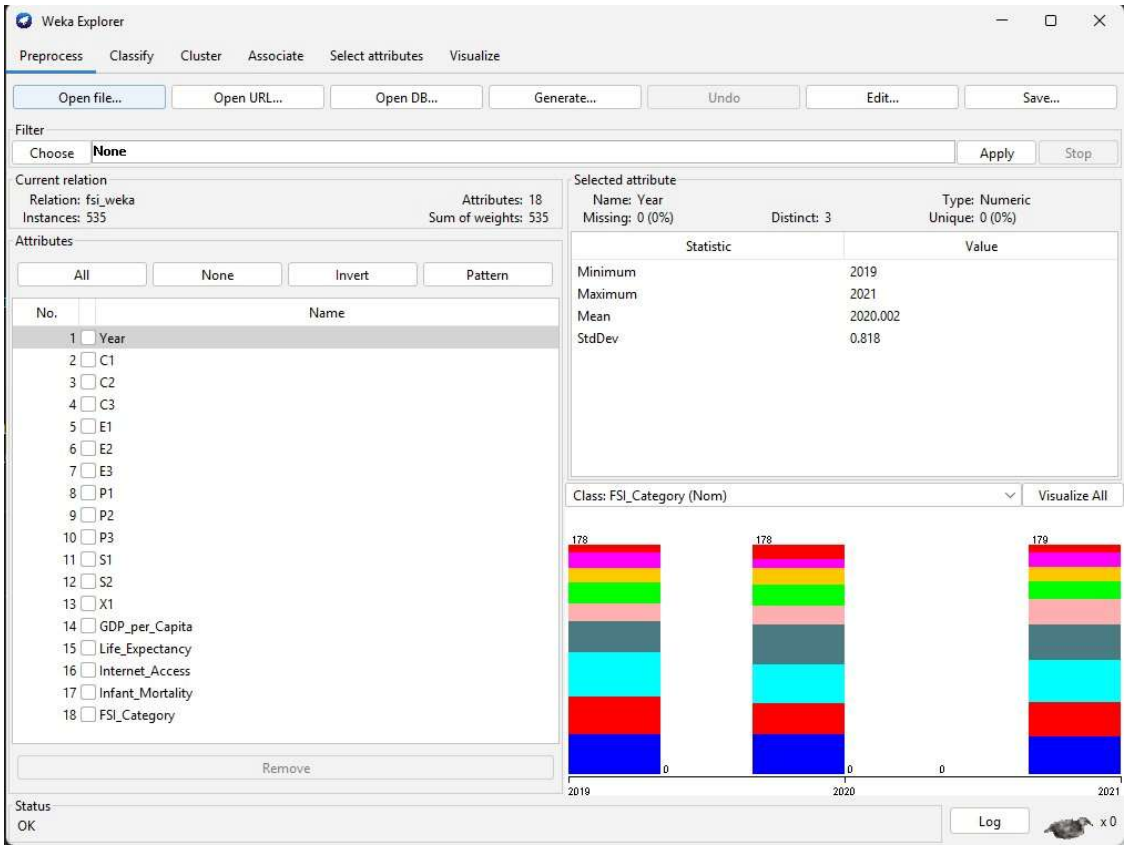

Figure 3: WEKA Preprocess tab showing fsi_weka loaded 535 instances, 18 attributes, FSI_Category as class attribute. Distribution chart shows all 3 years balanced.

Step 2: Selecting the Classify tab and choosing J48

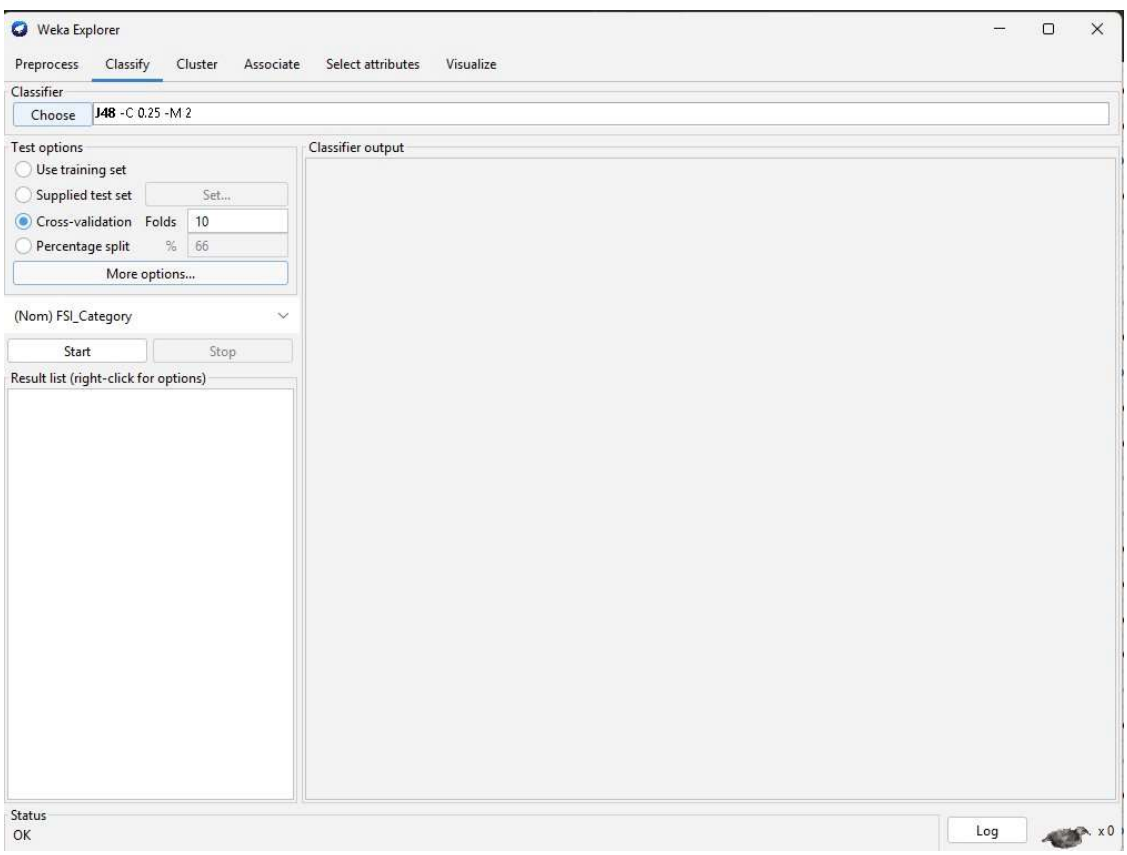

Figure 4: Classify tab with J48 selected as the classifier. 10-fold cross-validation with FSI_Category as the target class.

Step 3: J48 results accuracy and confusion matrix

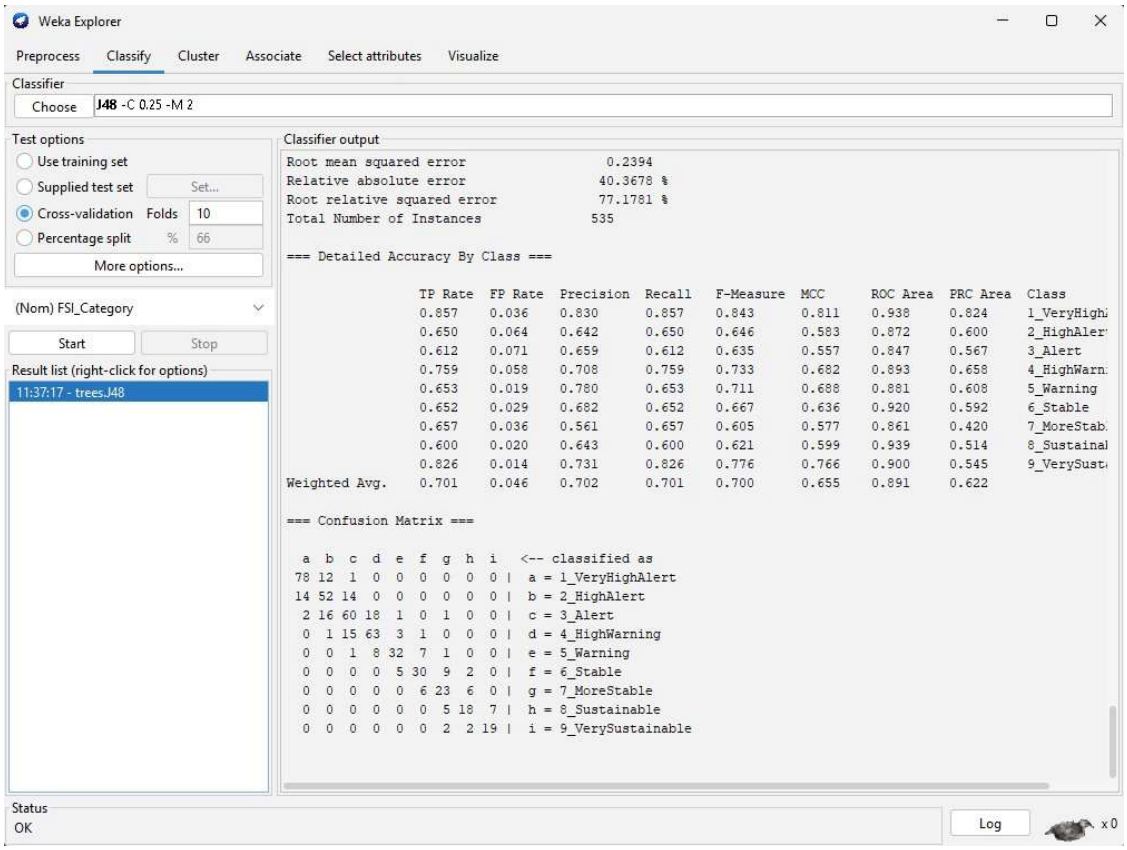

Figure 5: J48 classification results. Weighted average accuracy: 70.1%. Confusion matrix shows strong performance on 1_VeryHighAlert (TP=0.857) and 9_VerySustainable (TP=0.826).

Step 4: J48 decision tree rules (top section)

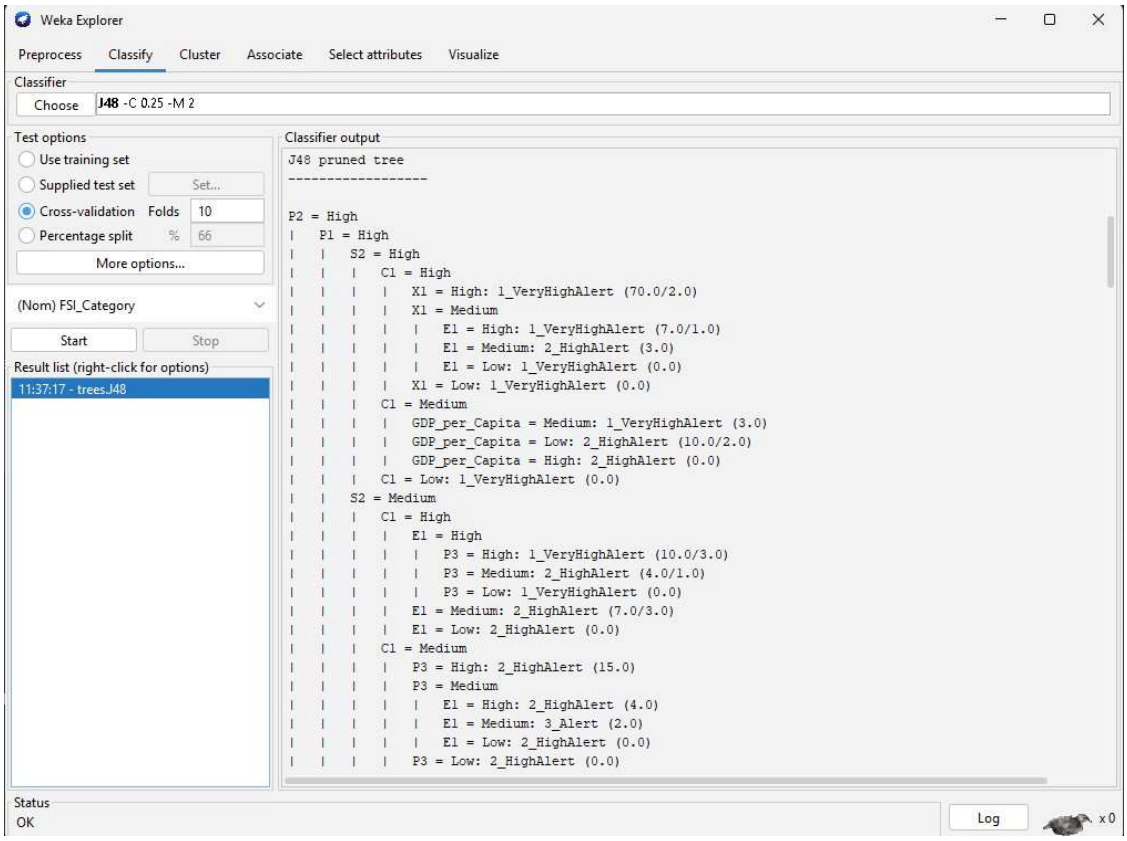

Figure 6: J48 pruned tree rules. Root split on P2 (Public Services). P2=High branch further splits on P1, S2, C1, X1 — classifying most fragile countries.

Step 5: J48 decision tree rules (continued)

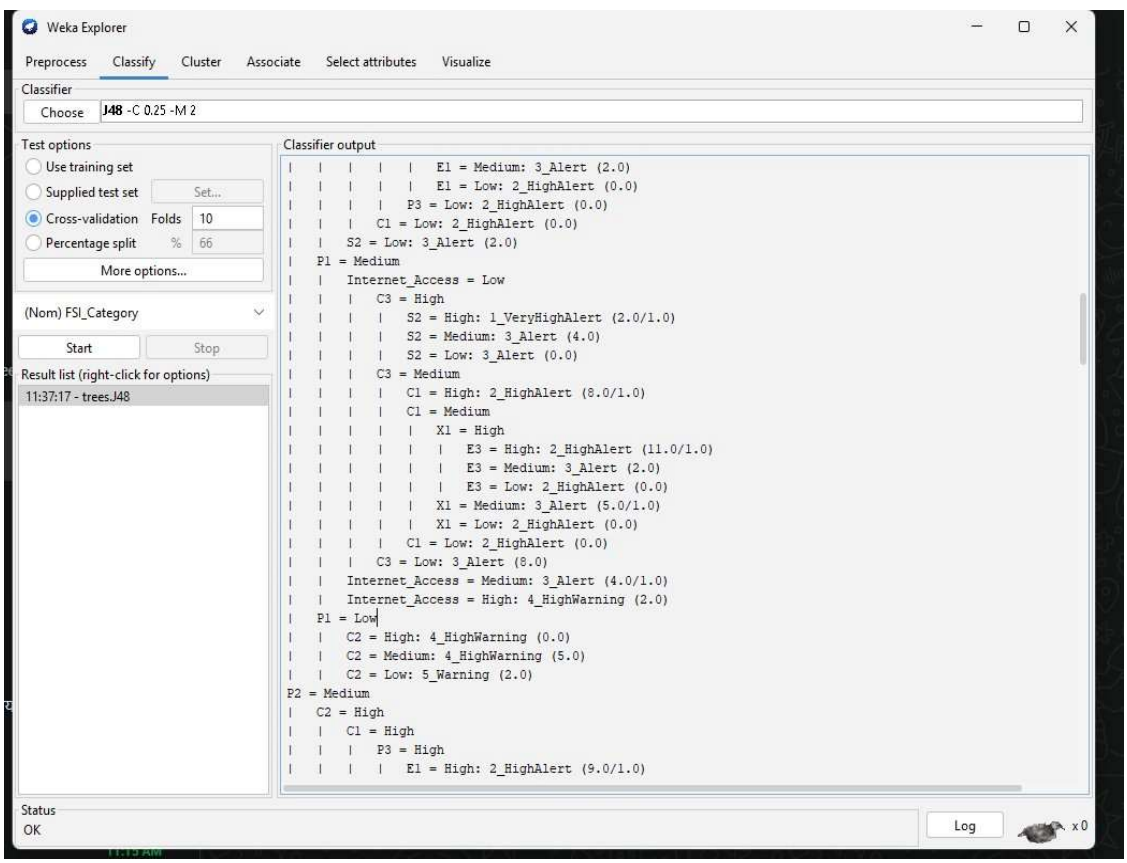

Figure 7: Continued J48 rules showing P2=Medium and P2=Low branches. Internet_Access, C2, GDP_per_Capita appear as secondary splits for less fragile categories.

Step 6: Decision tree visualization

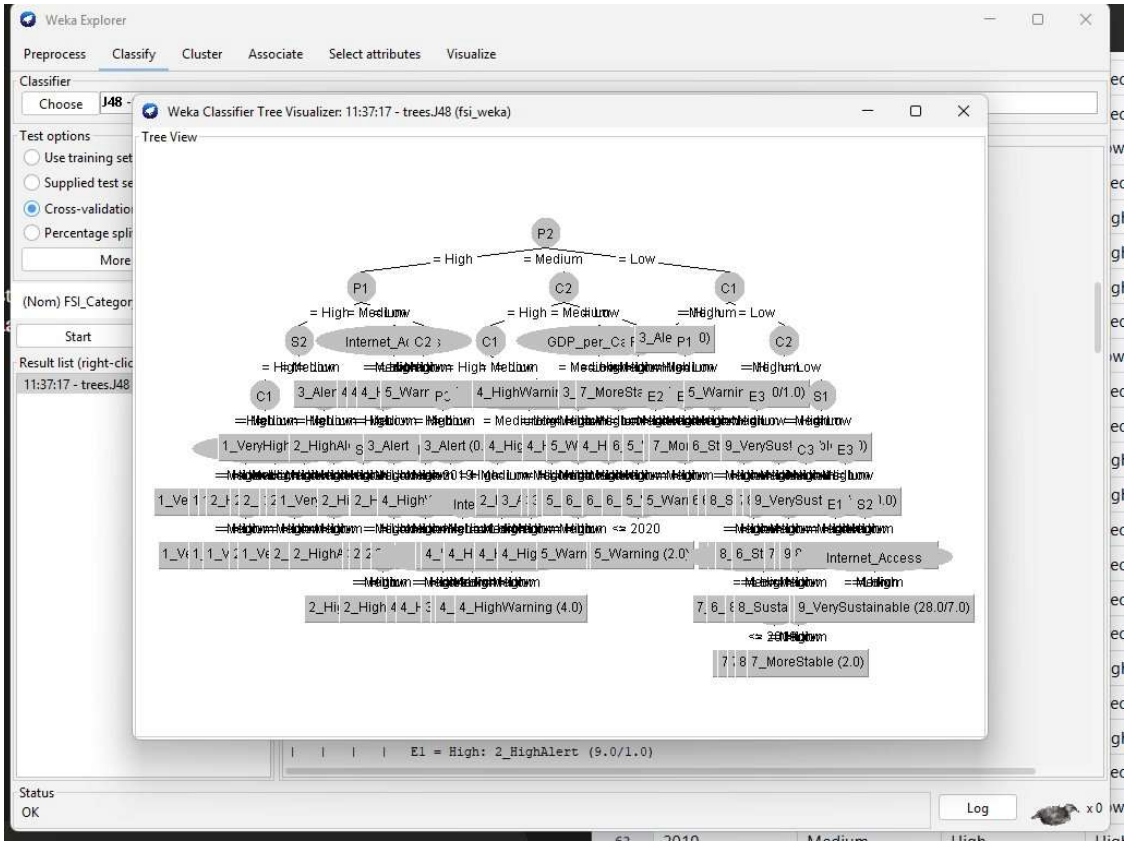

Figure 8: WEKA J48 tree visualization. Root node P2 branches into High/Medium/Low. P1, S2, C1, C2, GDP_per_Capita and Internet_Access are key internal nodes.

4.3 Key Rules from J48

Rule	Predicted Class
P2=High AND P1=High AND S2=High AND C1=High AND X1=High	1_VeryHighAlert
P2=High AND P1=High AND S2=Medium AND C1=Medium AND GDP_per_Capita=Low	2_HighAlert
P2=Medium AND C2=High AND C1=High AND P3=High AND E1=High	2_HighAlert
P2=Low AND C1=Low AND C2=Low AND Internet_Access=High	9_VerySustainable

The J48 tree confirms findings from the Python Decision Tree: P2 (Public Services) is the primary root split, consistent with it being the #2 feature by importance. The tree also highlights Internet_Access and GDP_per_Capita as key secondary splits, validating the World Bank feature enrichment decision.

5. Action Rule Mining LISp-Miner

5.1 Attribute Analysis: Stable vs. Flexible

Attribute	Type	Reason

Country	Stable	Identity cannot change
Year	Stable	Temporal identifier cannot change
C1, C2, C3, E1, E2, E3, P1, P2, P3, S1, S2, X1	Flexible	All 12 FSI sub-indicators can improve through policy
GDP_per_Capita, Life_Expectancy, Internet_Access, Infant_Mortality	Flexible	All 4 World Bank features can change through development
FSI_Category	Decision	Target goal is to move to a less fragile category

5.2 LISp-Miner Step-by-Step Screenshots

Step 1: Importing fsi_lisp_miner.csv

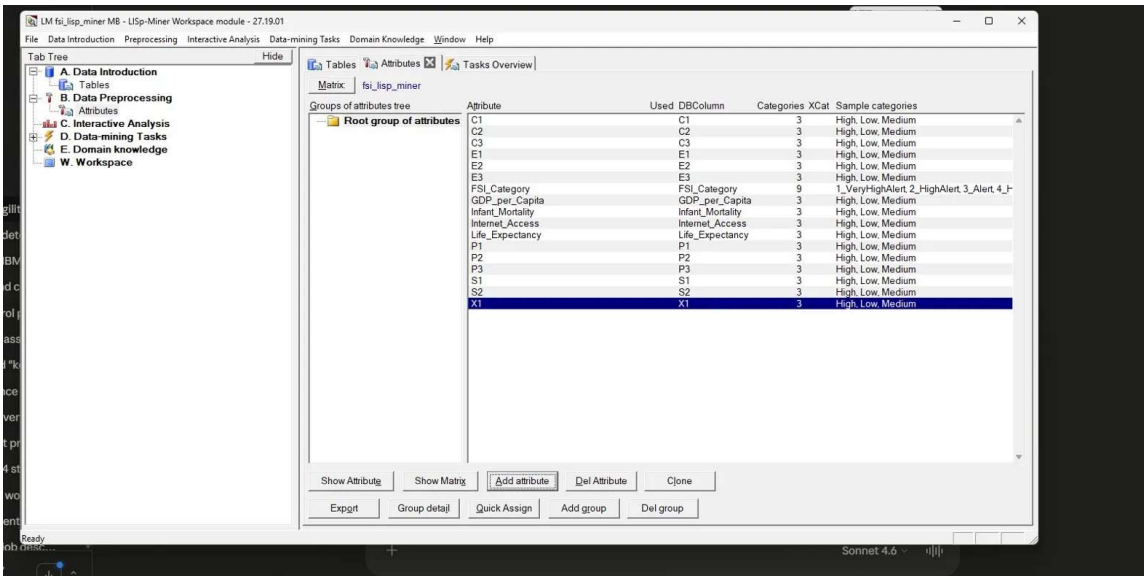

Figure 9: CSV import showing fsi_lisp_miner.csv loaded with all 19 columns, comma separator, 535 rows from C:\Users\vatsa\Downloads\KDD_proj.

Step 2: Table successfully loaded 535 rows

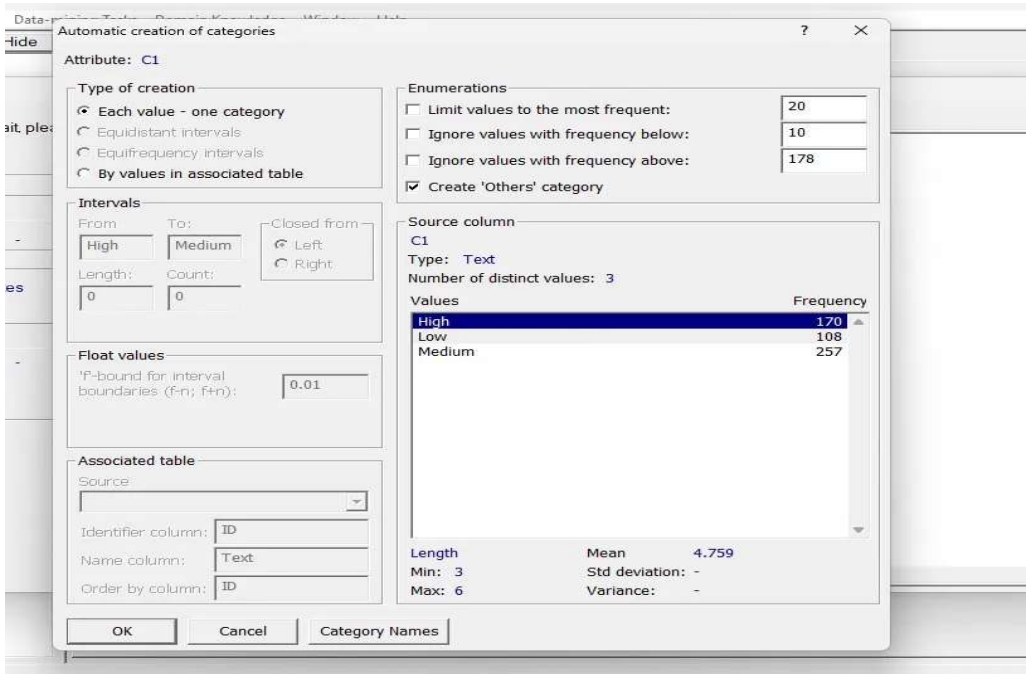

Figure 10: Tables view showing fsi_lisp_miner with 535 data rows, Valid = Yes.

Step 3: All database columns available

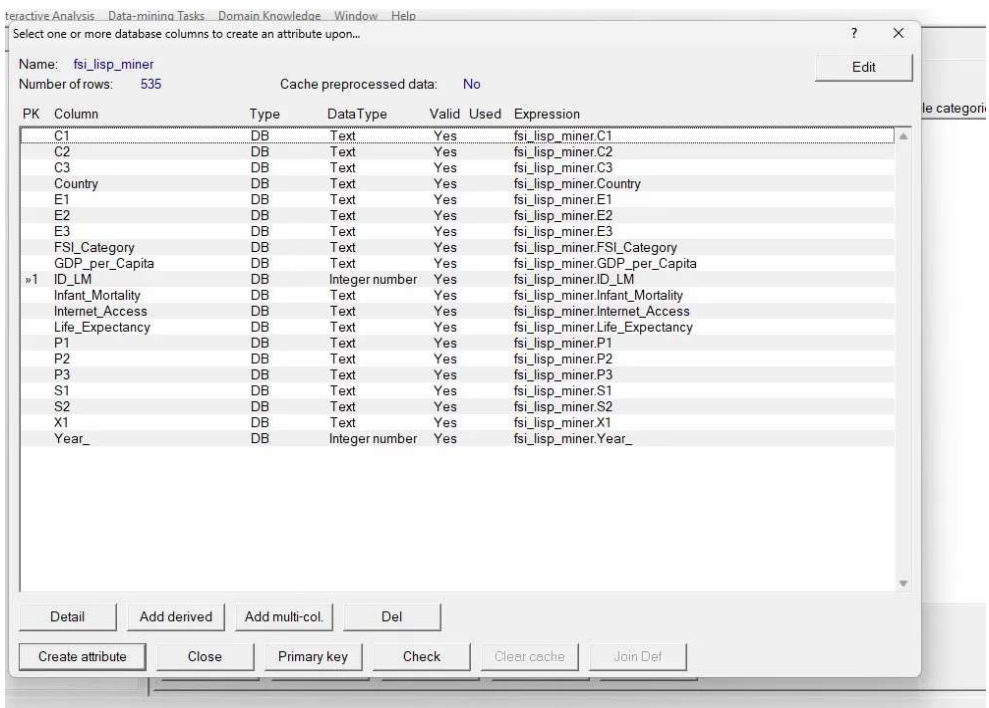

Figure 11: Column list showing all 19 columns including C1–X1, FSI_Category, GDP_per_Capita, Infant_Mortality, Internet_Access, Life_Expectancy.

Step 4: Creating categories for each attribute

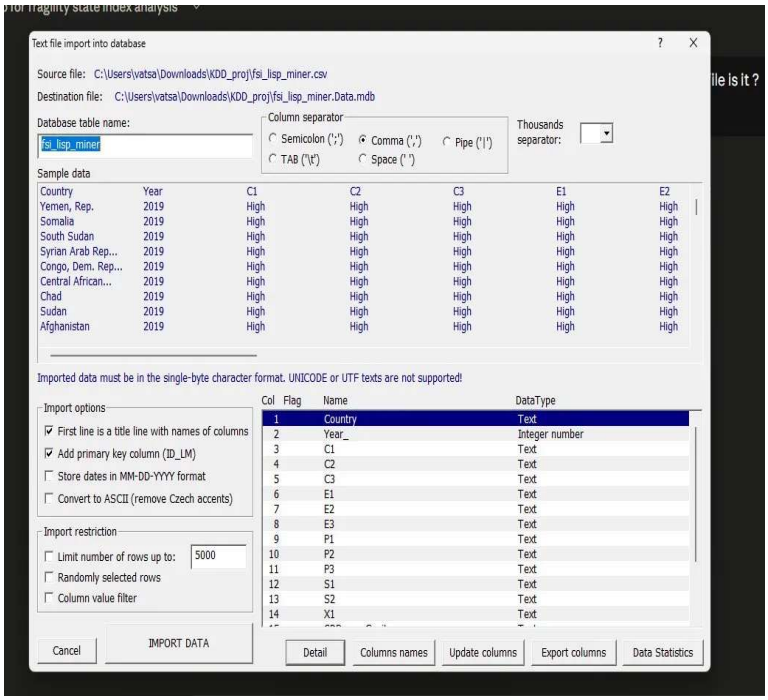

Figure 12: Category creation for C1 3 distinct values: High (170), Low (108), Medium (257). Each value = one category.

Step 5: All 17 attributes created

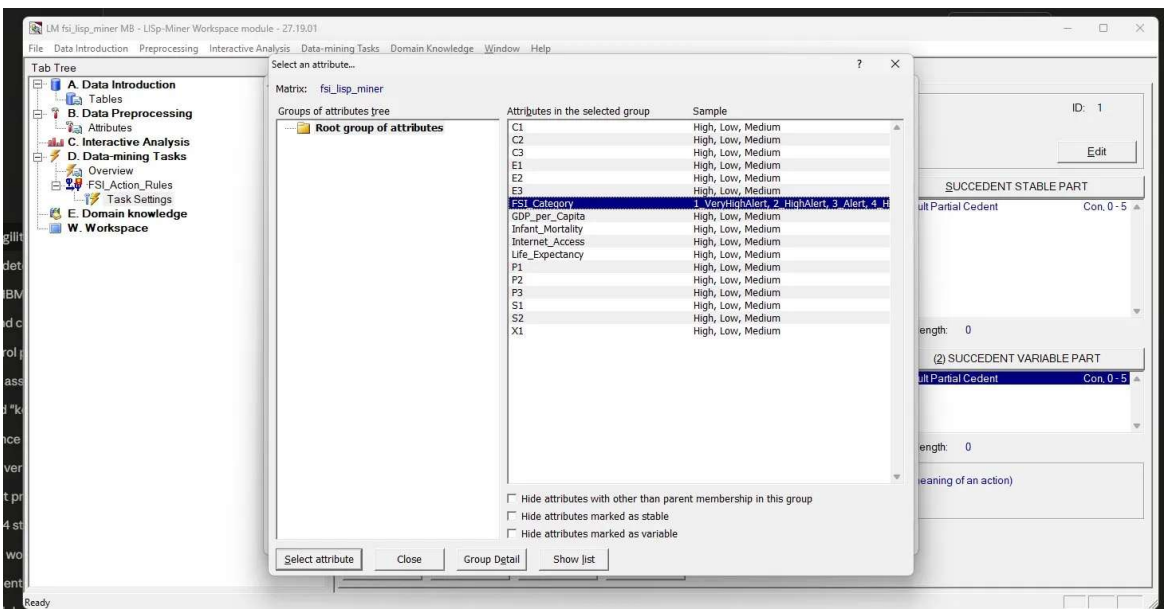

Figure 13: Attributes tab showing all 17 attributes. FSI_Category = 9 categories. All others = 3 categories (High, Low, Medium).

Step 6: Selecting Ac4ft-Miner task type

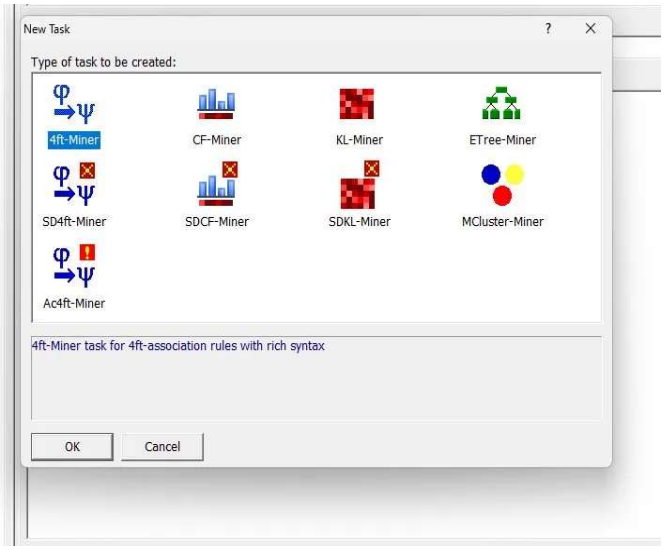

Figure 14: New Task dialog Ac4ft-Miner selected for action rule extraction.

Step 7: All attributes visible for assignment

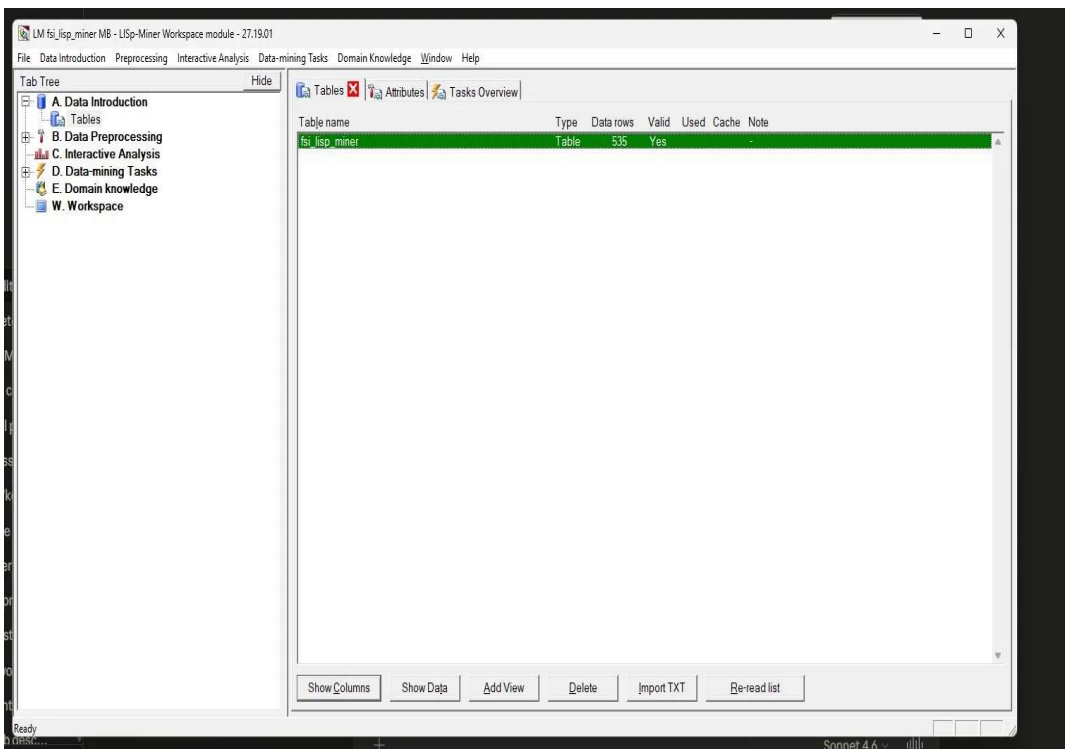

Figure 15: Attribute selector showing all 17 attributes. FSI_Category selected for Succedent Variable Part.

Step 8: Succedent FSI_Category 1_VeryHighAlert → 2_HighAlert

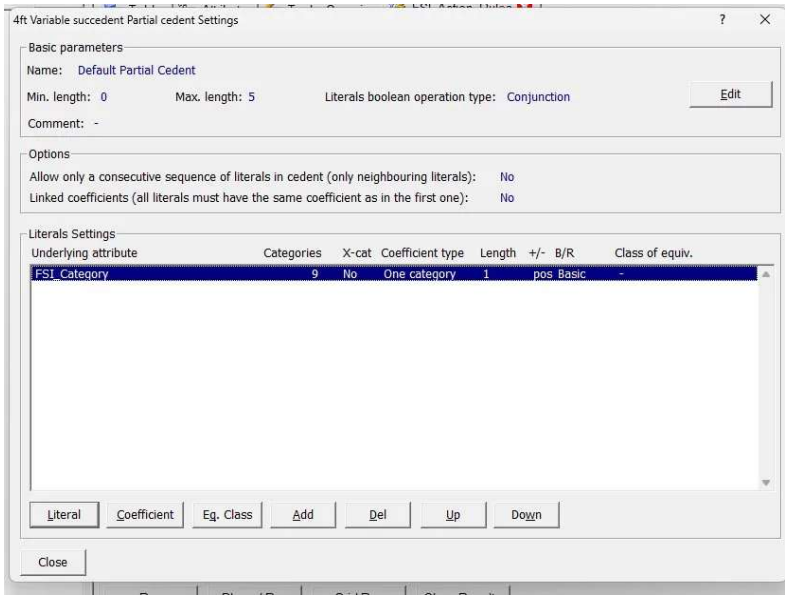

Figure 16: Succedent settings FSI_Category configured with One category coefficient, transition from 1_VeryHighAlert to 2_HighAlert.

Step 9: Antecedent P2, P1, C1, E1, GDP_per_Capita (High → Medium)

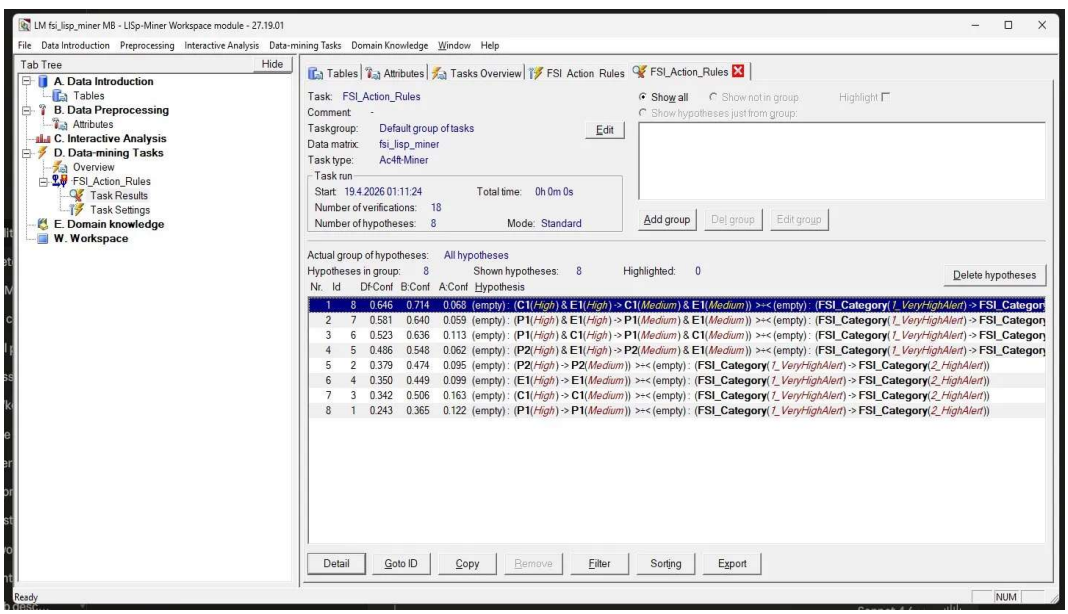

Figure 17: Antecedent Variable Part with 5 flexible attributes. Each set to One category, High → Medium transition representing improvement.

Step 10: Results 8 action rules discovered

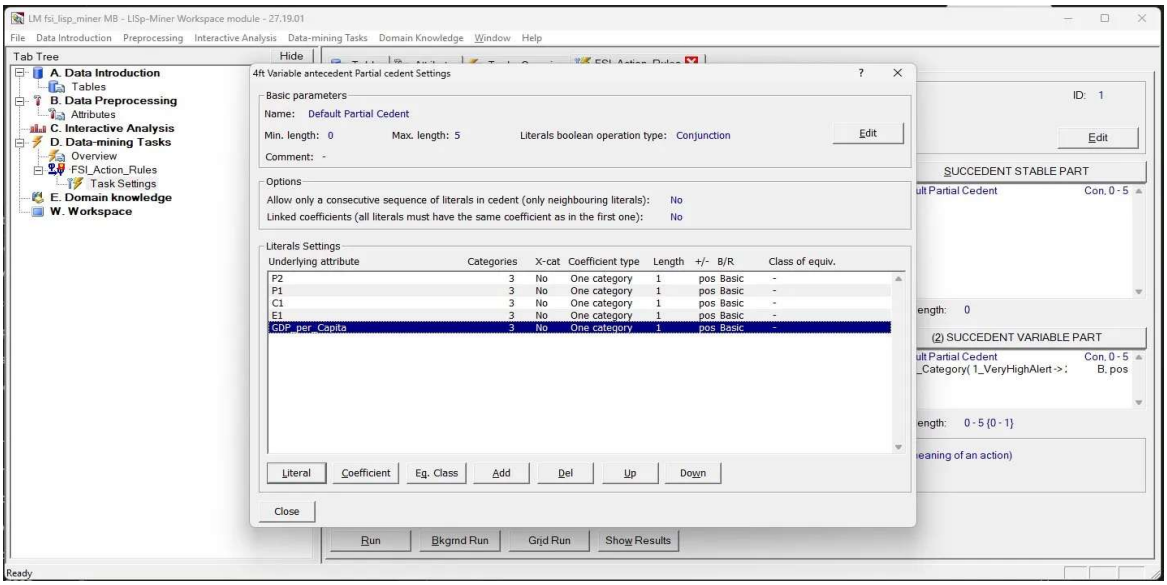

Figure 18: FSI_Action_Rules results 18 verifications, 8 hypotheses. Sorted by Df-Confidence with C1+E1 as strongest rule at 0.646.

5.3 Hypothesis Definition

The hypothesis was: if flexible attributes improve from High to Medium fragility, does FSI_Category transition from 1_VeryHighAlert to 2_HighAlert? Attributes P2, P1, C1, E1, and GDP_per_Capita were selected as antecedents because they are the top predictors from classification (feature importances). Minimum support was set to 5 absolute records for both before and after states.

5.4 Action Rules — All 8 Rules

Rank	Action Rule	Df-Conf	B:Conf	A:Conf
1	C1(High→Med) & E1(High→Med) ⇒ FSI(VeryHighAlert→2HighAlert)	0.646	0.714	0.068
2	P1(High→Med) & E1(High→Med) ⇒ FSI(VeryHighAlert→2HighAlert)	0.581	0.640	0.059
3	P1(High→Med) & C1(High→Med) ⇒ FSI(VeryHighAlert→2HighAlert)	0.523	0.636	0.113
4	P2(High→Med) & E1(High→Med) ⇒ FSI(VeryHighAlert→2HighAlert)	0.486	0.548	0.062
5	P2(High→Med) ⇒ FSI(VeryHighAlert→2HighAlert)	0.379	0.474	0.095
6	E1(High→Med) ⇒ FSI(VeryHighAlert→2HighAlert)	0.350	0.449	0.099
7	C1(High→Med) ⇒ FSI(VeryHighAlert→2HighAlert)	0.342	0.506	0.163
8	P1(High→Med) ⇒ FSI(VeryHighAlert→2HighAlert)	0.243	0.365	0.122

5.5 Analysis of Top Rules

Rule 1 Security Apparatus + Economy (Df-Conf: 0.646, strongest): If a country improves both C1 (Security Apparatus) and E1 (Economy) from High to Medium fragility, it is associated with moving from Very High Alert to High Alert. B:Confidence of 71.4% confirms the before-state is well anchored. The low A:Confidence (6.8%) shows these improvements are rare but highly directional. Applicable to South

Sudan and Chad where simultaneous security and economic reform is the primary intervention pathway.

Rule 2 State Legitimacy + Economy (Df-Conf: 0.581): Improving P1 (State Legitimacy) and E1 together is the second strongest pathway. Political reform paired with economic growth creates a compounding stability effect. Most applicable to Yemen, where governance credibility reform alongside economic stabilization is critical.

Rule 3 State Legitimacy + Security Apparatus (Df-Conf: 0.523): The third rule covers P1+C1 improvements. With 84 countries in the before-state and B:Confidence of 63.6%, this applies broadly across Very High Alert countries. Particularly relevant for post-conflict states under UN peacekeeping presence like Central African Republic.

Single-Factor Rules (Rules 5–8, Df-Conf: 0.24–0.38): All single-attribute rules show substantially lower confidence than two-factor combinations. This confirms fragility is multidimensional single-sector interventions alone are insufficient. Even the strongest single-factor rule (P2 alone at 0.379) is far weaker than any two-factor pair.

Country (2021)	C1	E1	P1	P2	Best Applicable Rule
South Sudan	High	High	High	High	Rule 1: C1+E1 security + economic reform
Yemen	High	High	High	High	Rule 2: P1+E1 governance + economic recovery
Central African Rep.	High	High	High	High	Rule 3: P1+C1 legitimacy + security reform

6. Key Findings

- Random Forest achieves best accuracy at 79.4% with enriched 16-feature dataset (+3.7% over original 12).
- WEKA J48 achieves 70.1% accuracy with 10-fold cross-validation, confirming P2 as the root decision node.
- GDP per Capita is the single most important predictor (importance ~0.207); Life Expectancy shows the strongest correlation with FSI (−0.793).
- All 4 World Bank features show |r| > 0.72 with FSI score all are meaningful additions.
- LISp-Miner extracted 8 action rules. The strongest (C1+E1, Df-Conf: 0.646) shows joint security and economic reform is the most impactful lever.
- Single-factor action rules (Df-Conf: 0.24–0.38) are much weaker than two-factor combinations, confirming fragility requires multi-sector interventions.

7. Tools and Technologies

Tool	Version	Purpose
Python (pandas, scikit-learn)	Python 3.10	Data preprocessing, classification, feature importance
World Bank REST API	worldbank.org	4 additional country-level features
Google Colab	Jupyter-based	Development environment
WEKA	v3.8.7	J48 decision tree classification and rule visualization
LISp-Miner	v27.19.01 Ac4ft-Miner	Action rule mining

matplotlib / seaborn	v3.7	Correlation and feature importance charts
GitHub	github.com/shruti-08	Version control

8. References

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- LISp-Miner. https://lispminer.vse.cz/ | Manual: https://webpages.uncc.edu/ras/Paper-AR.pdf
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- ITCS 6162 Knowledge Discovery in Databases. UNC Charlotte, Spring 2026.